

ORIGINAL ARTICLE

Tumor protein expression of the DNA repair gene BRCA1 and lethal prostate cancer

Konrad H.Stopsack^{1,2,*†,⊕}, Travis Gerke^{2,3,†}, Piotr Zareba^{4,5,†}, Andreas Pettersson^{2,6}, Dipanjan Chowdhury⁷, Ericka M.Ebot², Richard Flavin^{8,9}, Stephen Finn^{8,9}, Philip W.Kantoff¹, Meir J.Stampfer^{2,10}, Massimo Loda¹¹, Michelangelo Fiorentino^{2,12,‡} and Lorelei A.Mucci^{2,10,‡}

¹Department of Medicine, Memorial Sloan Kettering Cancer Center, New York, NY, USA, ²Department of Epidemiology, Harvard T.H. Chan School of Public Health, Boston, MA, USA, ³Department of Cancer Epidemiology, Moffitt Cancer Center, Tampa, FL, USA, ⁴Department of Surgery, Memorial Sloan Kettering Cancer Center, New York, NY, USA, ⁵Department of Urology, McMaster University, Hamilton, ON, USA, ⁶Department of Medicine, Clinical Epidemiology Unit, Solna, Karolinska Institutet, Stockholm, Sweden, ⁷Department of Radiation Oncology, Dana-Farber Cancer Institute, Harvard Medical School, Boston, MA, USA, ⁸Department of Pathology, St. James's Hospital, Dublin, Ireland, ⁹Trinity College, Dublin, Ireland, ¹⁰Channing Division of Network Medicine, Department of Medicine, Brigham and Women's Hospital, Harvard Medical School, Boston, MA, USA, ¹¹Department of Pathology, Cornell Medical School, New York, NY, USA and ¹²Pathology Unit, Addarii Institute, S. Orsola-Malpighi Hospital, Bologna, Italy

*To whom correspondence should be addressed. Department of Medicine, Memorial Sloan Kettering Cancer Center, 1275 York Avenue, New York, NY 10065, USA. Tel: 212-639-5851; Fax: 929-321-5023; Email: stopsack@mskcc.org.

†These authors share first authorship.

‡These authors share senior authorship.

Abstract

DNA repair genes are commonly altered in metastatic prostate cancer, but BRCA1 mutations are rare. Preliminary studies suggest that higher tumor expression of the BRCA1 protein may be associated with worse prognosis. We undertook a prospective study among men with prostate cancer in the Health Professionals Follow-up Study and evaluated BRCA1 via immunohistochemical staining on tissue microarrays. BRCA1 was expressed in 60 of 589 tumors. Prevalence of BRCA1 positivity was 43% in the 14 men with metastases at diagnosis compared with 9% in non-metastatic tumors [difference, 33 percentage points; 95% confidence interval (CI), 7–59]. BRCA1-positive tumors had 2.16-fold higher Ki-67 proliferative indices (95% CI, 1.18–3.95), higher tumor aneuploidy as predicted from whole-transcriptome profiling, and higher Gleason scores. Among the 575 patients with non-metastatic disease at diagnosis, we evaluated the association between BRCA1 expression and development of lethal disease (metastasis or cancer-specific death, 69 events) during long-term follow-up (median, 18.3 years). A potential weak association of BRCA1 positivity with lethal disease (hazard ratio, 1.61; 95% CI, 0.82–3.15) was attenuated when adjusting for age, Gleason score and clinical stage (hazard ratio, 1.11; 95% CI, 0.54–2.29). In summary, BRCA1 protein expression is a feature of more proliferative and more aneuploid prostate tumors and is more common in metastatic disease. While not well suited as a prognostic biomarker in primary prostate cancer, BRCA1 protein expression may be most relevant in advanced disease.

Abbreviations

CI	confidence interval
HPFS	Health Professionals Follow-up Study
TUNEL	Terminal Deoxynucleotidyl Transferase-Mediated dUTP Nick End Labeling

Introduction

The BRCA1 protein is a component of the homologous recombination pathway that repairs DNA double strand breaks to maintain genomic integrity (1,2). BRCA1 may additionally facilitate DNA repair by regulating DNA replication, cell cycle checkpoint control, chromatin remodeling and mitotic spindle pole assembly. Failure to repair damaged DNA can trigger apoptosis or produce mutations that promote carcinogenesis.

BRCA1 is classically regarded as a tumor suppressor gene. Germline mutations in BRCA1 that result in an absent or defective BRCA1 protein have been associated with an increased risk of prostate cancer (3–8). However, both germline and tumor mutations (9) in BRCA1 are rare even in men with metastatic prostate cancer, compared with BRCA2 mutations. Mutations in either BRCA1 or BRCA2 (analyzed combined) are associated with a higher risk of metastasis and death from prostate cancer (7,8,10–12), but it is unclear whether these data apply to BRCA1 mutations by themselves.

Paradoxically, immunohistochemical expression of BRCA1 protein appears higher in prostate cancer than benign prostatic epithelium (13,14). Increased tumor expression of the BRCA1 protein has also been associated with worse survival in patients with ovarian cancer (15–18) and breast cancer (19). BRCA1 protein overexpression may either promote tumor progression, for example by inducing chromosomal instability and aneuploidy (2,20), or alternatively may be expressed in response to malignant processes that promote DNA damage, including tumor proliferation. In a study of 393 participants of the Physicians' Health Study with prostate cancer, our group previously found nuclear BRCA1 protein expression to be present in 15% of clinically localized prostate tumors, to be associated with an increased Ki-67 tumor proliferation index, and to be associated with higher rates of metastasis and prostate cancer-specific mortality after adjusting for clinical factors (13).

The objective of the current study was to evaluate the association between BRCA1 protein expression and features of prostate cancer aggressiveness in a large independent cohort of men followed prospectively for development of lethal prostate cancer.

Patients and methods

Study population

The Health Professionals Follow-up Study (HPFS) enrolled 51 529 male dentists, pharmacists, optometrists, osteopaths, podiatrists and veterinarians between 40 and 75 years of age in 1986. Study participants are asked to complete a questionnaire relating to their lifestyle habits and medical history every 2 years. The current study included men diagnosed with prostate cancer during prospective follow-up of HPFS with available surgical tissue between 1986 and 2000. Tumors diagnosed via transurethral resection of the prostate were not excluded.

Dates of cancer diagnosis, stage and prostate-specific antigen level at diagnosis were abstracted from medical records and pathology reports. Dates and causes of death were assigned by way of a centralized review of medical records and death certificates by study physicians. Since 2000, data on development of distant metastases have additionally been tracked by way of patient questionnaires and review of medical records.

The study was conducted in accordance with the U.S. Common Rule and approved by the institutional review boards of Partners Healthcare and Harvard T.H. Chan School of Public Health, and those of participating registries as required.

Pathology and immunohistochemistry

We retrieved the archival formalin-fixed, paraffin-embedded tumor tissue specimens from men in the cohort diagnosed with prostate cancer. Hematoxylin–eosin slides were centrally re-reviewed for Gleason grade and other histologic features by expert genitourinary pathologists (M.F., R.F., S.F. and M.L.) (21). Slides were also reviewed to identify areas of high-density tumor for construction of tumor tissue microarrays. At least three tumor cores (0.6 mm) were sampled from each tumor, using a manual arrayer; five tissue microarrays were used in this study.

Immunohistochemical staining for BRCA1 was performed on 5 μ m tumor sections of each tissue microarray as described (13). Briefly, after ethylenediaminetetraacetic acid-based antigen retrieval, a monoclonal antibody for the N-terminal of the functional BRCA1 protein was used (clone MS110; Calbiochem) at dilution 1:50. Nuclear BRCA1 tumor expression was scored manually by the study pathologists as presence or absence of any staining, regardless of staining intensity.

Tissue microarrays were also stained for the proliferation marker Ki-67 and scored by quantitative image analysis as the number of Ki-67-positive nuclei over the total number of neoplastic nuclei, as described previously (22). A Terminal Deoxynucleotidyl Transferase-Mediated dUTP Nick End Labeling (TUNEL) assay was used to assess apoptosis and the apoptotic rate was scored semi-quantitatively by eye, as described previously (23). The presence of the most common molecular prostate cancer subtype defined by the TMPRSS2:ERG fusion was determined by immunohistochemical staining for ERG protein, as described previously (24). Whole-transcriptome gene expression profiling data were used to predict the number of altered (gained or deleted) chromosome arms as a measure of tumor aneuploidy, as described previously (25).

Table 1. Participant and tumor characteristics by BRCA1 tumor protein expression, HPFS, 1986–2000

	BRCA1-negative	BRCA1-positive
Participants/tumors, n	529	60
Age at cancer diagnosis [years], median (interquartile range)	66 (62–70)	66 (63–69)
Gleason score, n (%)		
5–6	96 (18%)	6 (10%)
3 + 4	200 (37%)	17 (28%)
4 + 3	134 (25%)	20 (33%)
8	36 (7%)	6 (10%)
9–10	63 (12%)	11 (18%)
Clinical TNM stage, n (%)		
T1/T2	457 (90%)	47 (80%)
T3	35 (7%)	3 (5%)
T4/N1	8 (2%)	3 (5%)
M1	8 (2%)	6 (10%)
(missing)	21	1
Pathological TNM stage, n (%)		
T2	343 (69%)	36 (74%)
T3a	98 (20%)	7 (14%)
T3b	41 (8%)	5 (10%)
T4/N1	17 (3%)	0 (0%)
(missing)	30	12
Prostate-specific antigen (ng/ml), n (%)		
<4	53 (13%)	9 (22%)
4–10	257 (63%)	19 (48%)
10–20	93 (23%)	12 (30%)
>20	52 (11%)	12 (23%)
(missing)	74	8
ERG-positive, n (%)	260 (51%)	27 (45%)

Statistical analysis

In a cross-sectional analysis at cancer diagnosis, differences in BRCA1 positivity according to ERG status and stage at diagnosis (metastatic/M1 versus non-M1) were estimated using binomial regression models analogous to the approach of Spiegelman and Hertzmark (26). The proliferative index Ki-67 was quantile-normalized between tissue microarrays to account for batch effects of automated scoring. Ki-67, TUNEL and the number of altered chromosome arms as a measure of aneuploidy were used as outcomes of linear regression models; Ki-67 and TUNEL were transformed using a natural logarithm to obtain expression ratios. Models for Ki-67, TUNEL and aneuploidy indices were additionally adjusted for Gleason score (categorical: <7, 3 + 4, 4 + 3, 8, 9–10).

In a longitudinal analysis among men diagnosed with non-M1 prostate cancer, associations between BRCA1 positivity at diagnosis and development of lethal prostate cancer (distant metastases or death due to prostate cancer) were assessed using Cox proportional hazards models. Person-time was calculated from the date of diagnosis to development of metastases or death due to prostate cancer (if the date of metastasis was unknown), with censoring at death from other causes or end of follow-up for this analysis in January 2016. Multivariable models included age at diagnosis (continuous), Gleason score (as above) and clinical stage (categorical: T1/T2; T3; T4/N1; with mode imputation for missing values).

Results

Among the 589 men in the study, the median age at cancer diagnosis was 66 years (interquartile range, 62–70). Clinical stage was T1/T2 in 504 patients (86%); 14 patients (2%) had M1 disease at diagnosis. The majority of tissue were prostatectomy specimens ($n = 550$; 93%), with the remainder from transurethral resection of the prostate. Data for additional biomarkers were available for ERG ($n = 574$), Ki-67 ($n = 539$), TUNEL ($n = 451$) and the number of altered chromosome arms predicted from transcriptome profiling ($n = 199$).

Overall, BRCA1 protein expression was detectable by immunohistochemistry in 60 patients [10% of the study population; 95% confidence interval (CI), 8–13]. BRCA1 positivity was associated with both higher grade and advanced stage (Table 1). Among non-metastatic tumors, 9% were positive for BRCA1 (54 of 576), while 43% of primary prostate tissues in the setting of metastatic disease at diagnosis were positive for BRCA1 (6 of 14; prevalence difference, 33 percentage points; 95% CI, 7–59). More tumors with Gleason scores 8–10 had BRCA1 positivity than

those with Gleason scores 5–6 (prevalence difference, 9 percentage points; 95% CI, 1–17). Between ERG-negative and ERG-positive tumors, prevalence of BRCA1 positivity did not differ appreciably (difference, –2 percentage points; 95% CI, –7 to 3).

BRCA1-positive tumors had a 2.16-fold higher proliferation index Ki-67 (95% CI, 1.18–3.95) compared with BRCA1-negative tumors (Figure 1A). The association between BRCA1 positivity and the apoptosis marker TUNEL was weaker, with 1.43-fold higher TUNEL (95% CI, 0.97–2.13) in BRCA1-positive tumors compared with BRCA1-negative tumors (Figure 1B). BRCA1-positive tumors had higher levels of aneuploidy, with a mean difference of 1.5 additional predicted altered chromosome arms compared with BRCA1-negative tumors (95% CI, 0.1–2.8; Figure 1C). When adjusting for Gleason grade, associations of BRCA1 with Ki-67 (1.96-fold difference between BRCA1-positive and BRCA1-negative; 95% CI, 1.07–3.59) and TUNEL (1.42-fold; 95% CI, 0.95–2.11) were only minimally changed, while the association with aneuploidy indices was somewhat attenuated (mean difference, 1.1 chromosome arms; 95% CI, –0.2 to 2.5).

During 9362 person-years of follow-up (median, 18.3 years), 69 of the 575 men diagnosed initially with non-metastatic disease developed distant metastases or died of prostate cancer (lethal disease; Table 2). The cumulative incidence of lethal disease over the entire 29 years of follow-up was 30% in BRCA1-positive tumors and 24% in BRCA1-negative tumors (Figure 2). In unadjusted models, there was a weakly positive association between BRCA1 positivity and lethal disease (hazard ratio, 1.61; 95% CI, 0.82–3.15). The association was further attenuated (hazard ratio, 1.11; 95% CI, 0.54–2.29) upon adjusting for age at cancer diagnosis, Gleason score and clinical stage.

Discussion

This prospective cohort of men with prostate cancer provided evidence that tumors expressing the BRCA1 protein have a more aggressive phenotype than BRCA1-negative tumors. In our study, these tumors had higher Gleason scores, increased indices of tumor proliferation, more aneuploidy and possibly increased apoptotic indices. BRCA1 protein expression was substantially more common in tumors that were metastatic at diagnosis. Over three decades of prospective follow-up for metastases and death from prostate cancer, BRCA1 expression in

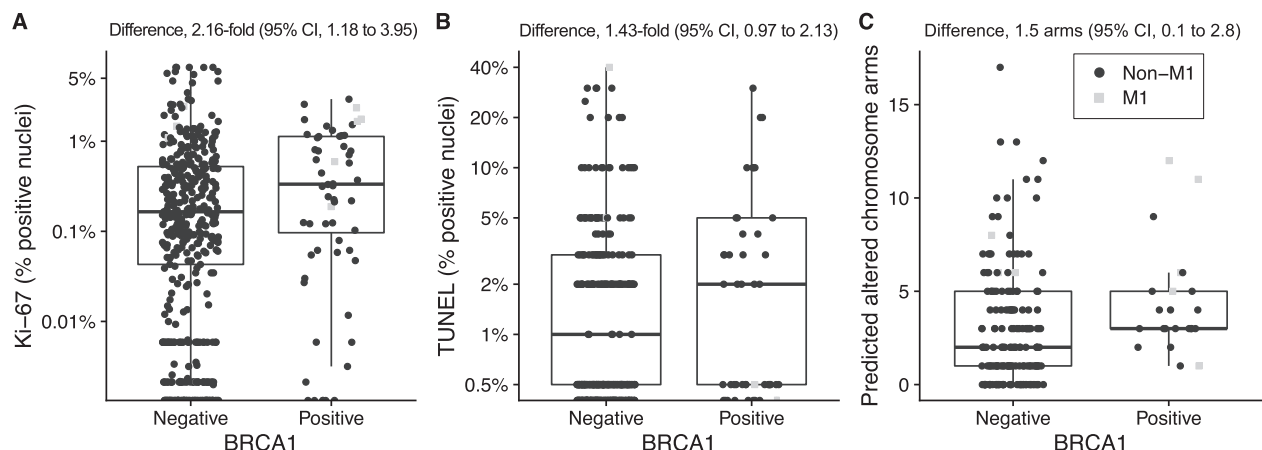
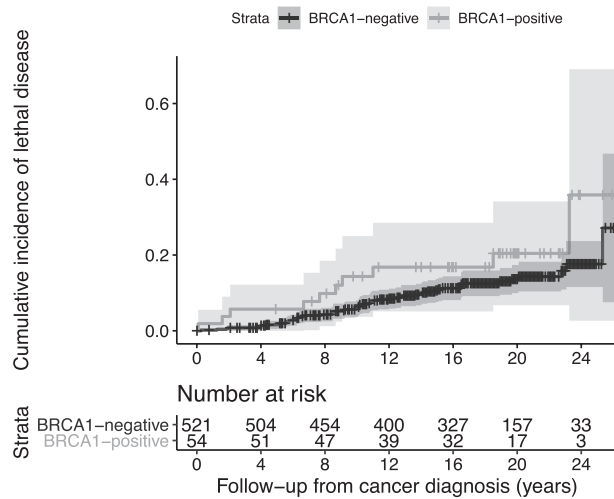


Figure 1. BRCA1 tumor protein expression and the Ki-67 proliferative index (A), the TUNEL apoptotic index (B) and the aneuploidy score, i.e. the number of altered (gained or lost) chromosome arms predicted from transcriptome profiling (C). Symbols indicate stage M1 versus non-M1; y-axes are natural logarithm-scaled (A and B); boxes extend from the 25th to the 75th percentile, with thick lines indicating the median and whiskers extending 1.5 times the interquartile range.

Table 2. BRCA1 tumor protein expression and progression to lethal prostate cancer among men with primary, non-metastatic prostate cancer in the HPFS, 1986–2016

	BRCA1-negative	BRCA1-positive
Patients, n	521	54
Person-years	8484	878
Lethal events, n	59	10
Rate of lethal disease, per 1000 person-years	7.0	11.4
Risk of lethal disease, % (95% CI) ^a	24 (12–43)	30 (13–60)
Hazard ratio for lethal disease (95% CI)		
Unadjusted	1 (reference)	1.61 (0.82–3.15)
Adjusted for age, Gleason score, clinical stage	1 (reference)	1.11 (0.54–2.29)

^aPer Kaplan–Meier estimator over the entire follow-up (29 years).

**Figure 2.** BRCA1 tumor protein expression and cumulative incidence of lethal disease among 575 men with primary, non-metastatic prostate cancer in the HPFS, 1986–2016.

non-metastatic tumors was potentially modestly associated with higher rates of progression to metastases and prostate cancer death, although this association was null when adjusting for clinical factors.

Our results validate that BRCA1 expression is a feature of more aggressive tumors with higher grade, stage and proliferative indices, consistent with our prior results in the Physicians' Health Study. In contrast, associations of BRCA1 and lethal disease in this study are considerably weaker than in our prior study, which had a mean follow-up of 10.6 years and showed tumor BRCA1 protein expression to be a strong predictor of lethal prostate cancer after adjusting for age at cancer diagnosis and Gleason grade (hazard ratio, 2.5; 95% CI, 1.3–4.8) (13). Potential reasons for differences in results for lethal disease, while difficult to delineate definitively, may be related to secular changes in how prostate cancer is detected and staged, with fewer tumors in the Physicians' Health Study being diagnosed following prostate-specific antigen screening. We also note that numbers of lethal events among men with BRCA1-positive tumors in either study were limited, even though results in this current study are based on 9362 person-years of follow-up for lethal disease.

Our results highlight a potentially complex and opposing role of the DNA repair pathway in prostate cancer. Protein tumor expression of BRCA1 may have different implications and consequences on tumor biology than tumor or germline DNA mutations in BRCA1. The positive association between BRCA1

expression and the proliferative index Ki-67 in this study and our previous study raises the possibility that BRCA1 protein expression may be upregulated in highly proliferative tumors as a reaction to the increased DNA damage. This hypothesis is aligned with our observation that extent of tumor apoptosis may be higher in BRCA1-positive tumors. Rapid replication and growth of cancer cells, in the setting of insufficient telomerase activity and mutations in other DNA damage response pathways, results in a rapid accrual of double strand breaks that, if not promptly repaired, trigger cancer cell death through apoptosis (27,28). Overexpression of BRCA1 protein and other homologous repair pathway components could therefore be an adaptive mechanism that confers a survival advantage to the cancer cell by averting double strand break-induced apoptosis. Concomitantly, our results suggest that BRCA1 overexpression may result in aneuploidy, a feature of more aggressive prostate tumors (25), as also demonstrated in experimental models in breast cancer (20). It is less likely that BRCA1 overexpression is in turn due to aneuploidy via BRCA1 copy number gains on chromosome 17q, because BRCA1 DNA copy numbers have not been found to be tightly associated with BRCA1 protein expression in prostate cancer (29).

Whether BRCA1 protein expression status may have therapeutic implications presents an avenue for exploration. Patients with breast and ovarian cancers who have BRCA1 germline mutations are more likely to benefit from therapies with poly-ADP(ribose) polymerase-1 inhibition (30). In metastatic prostate cancer, germline and tumor mutations in BRCA1 are rare, precluding firm inferences (31–33). Clinical and preclinical studies also suggest that BRCA1 mutational status and BRCA1 expression levels may modulate chemosensitivity to a number of DNA-damaging agents including platinum-based chemotherapy (34,35). These observations beg the question whether prostate cancers differ in sensitivity to chemotherapy, radiation therapy and poly-ADP(ribose) polymerase-1 inhibitors according to BRCA1 protein status.

In summary, BRCA1 protein was expressed in 10% of primary prostate cancers in this largest study of BRCA1 protein expression in prostate cancer to date. BRCA1-positive tumors had higher Gleason scores, higher proliferative indices and more aneuploidy. BRCA1 was more frequently expressed in primary prostate cancer tissue in the setting of metastatic disease. Although BRCA1 expression does not appear to be a promising prognostic biomarker in primary prostate cancer, our findings, together with the intimate role that BRCA1 plays in modulating response to tumor DNA-directed therapies, suggest that BRCA1 protein expression should be considered for testing as a predictive biomarker for benefit from commonly used therapies for advanced prostate cancer.

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Conflict of Interest Statement: P.W. Kantoff reports investment interest in Context Therapeutics LLC, DRGT, Placon, Seer Biosciences and Tarveda Therapeutics, he is a company board member for Context Therapeutics LLC and is a consultant/scientific advisory board member for Bavarian Nordic Immunotherapeutics, DRGT, GE Healthcare, Janssen, New England Research Institutes, OncoCellMDX, Progenity, Sanofi, Seer Biosciences, Tarveda Therapeutics and Thermo Fisher, and serves on data safety monitoring boards for Genentech/Roche and Merck.

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